Reg. No. | | | | | | | | | | | [18MTEEM244]

SECOND SEMESTER M.TECH EXAMINATION, JUNE - 2019

ENERGY MANAGEMENT

Time: 03 Hours **Maximum Marks:100**

Instructions: **i. Answer FIVE full Questions. Question No. 1 and 2 are compulsory.**
ii. Missing data may be suitably assumed.

1.	**a.**	Discuss briefly on mechanism of ozone layer depletion and its impact on environment.	**10**
	b.	Explain the features of energy policy.	**10**
2.	**a.**	What are energy conservation opportunities? List ten points for energy conservation in steam and condensate system.	**10**
	b.	Write brief notes on different types of cogeneration plants.	**10**
3.	**a.**	Determine the balance which will accumulate at the end of year 4 in an account which pays 10% per year if a deposit of Rs.50000 is made today based on-	**06**
		i. Simple interest	
		ii. Compound interest	
	b.	To accumulate Rs.100000 for five years from today in an account earning 8% every year compound interest, how much must be deposited today?	**06**
	c.	Write short notes on payback period and internal rate of return method of project evaluation.	**08**
		OR	
4.	**a.**	Enumerate and briefly explain the factors affecting the capital budgeting.	**10**
	b.	Enumerate the steps involved in capital budgeting.	**10**
5.	**a.**	Enlist different types of recuperators and explain the causes of failure of a metallic recuperator.	**10**
	b.	Describe the working of a metallic recuperator with a neat sketch.	**10**
		OR	
6.	**a.**	How do you plan and conduct a waste heat survey? Discuss.	**10**
	b.	Explain in detail the economics for waste heat recovery.	**10**
7.	**a.**	What is load factor? What is its significance? How can it be improved?	**08**
	b.	What is load curve? What is the significance of load curve? What are the benefits of load curve analysis?	**12**
		OR	
8.	**a.**	Discuss on energy efficiency improvement opportunities in motors.	**10**
	b.	Write short notes on:	**10**
		i. Energy costs.	
		ii. Two part tariff.	

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